PROGRAME STRUCTURE				
RSCOM B.Sc. AVE				
1st Semester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Papers				
1	AVE092M101	History of Animation and Multimedia	100	3
2	AVE092M112	Fundamentals of Drawing for Animation	100	3
Minor Papers				
3	AVE092N113	Foundation of Colour Theory	100	3
Skill Enhancement Courses (SEC-1)				
4	AVE092S111	Creative Studio I (Characters & Life Studies)	100	3
Value Added Course (VAC-1)				
5	VAC992V1409	Introduction to Speculative Design	100	3
Interdisciplinary Course (IDC-1)				
6	IKS992I101	Indian Knowledge System I	100	3
Ability Enhancement Course (AEC-1)				
7	CEN982A101	CEN IV: Business Communication: Introduction to Effective Communication	100	2
Total Credit for 1st Semester				**20**
2ndSemester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Papers				
1	AVE092M211	Script Writing and Storyboarding	100	3
2	AVE092M212	Principles of Animation	100	3
Minor Papers				
3	AVE092N211	Commercial Art	100	3
Skill Enhancement Courses (SEC-2)				
4	AVE092S211	Creative Studio II (Layout & Background Design)	100	3
Value Added Course (VAC-2)				
5			100	3
Interdisciplinary Course				
6	IKS992I201	Indian Knowledge System II	100	3
Ability Enhancement Course (AEC-2)				
7	CEN982A201	CEN II : Approaches to Verbal and Non-Verbal Communication	100	2
Total Credit for 2nd Semester				**20**
Conferring the Certificate in Animation & Visual Effects, (CAVE)				

3rd Semester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M311	Tradigital Animation	200	4
2	AVE092M312	Introduction to 3D Modelling	200	4
Minor Papers				
3	AVE092N311	Stopmotion and Cutout Animation	200	4
Skill Enhancement Courses (SEC-3)				
4	AVE092S311	Creative Studio III (Design Thinking Process)	200	3
Interdisciplinary Course				
5	AVE092I311	Traditional Art and Contemporary Design Integration	200	3
Ability Enhancement Course (AEC-3)				
6	CEN982A301	CEN III : Approaches to Verbal and Non - Verbal Communication	200	1
7	BHS982A302	Behavioural Science – III	200	1
Total Credit for 3rd Semester				20

4th Semester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M411	Fundamentals of Texturing	200	4
2	AVE092M412	Introduction to Rigging	200	4
3	AVE092M413	Creative Studio IV (Indian Culture and Heritage Preservation) **(IKS based)**	200	4
Minor Papers				
4	AVE092N411	Fundamentals of Audio Video Editing	200	3
5	AVE092N412	Basics of 2D Animation	200	3
Ability Enhancement Course (AEC)				
6	CEN982A401	CEN IV : Business Communication : Concepts and Skills	200	2
Total Credit for 4th Semester				20

5th Semester				
Sl. No.	**Subject Code**	**Names of Subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M511	CGI Integration	300	4
2	AVE092M512	Lighting for Animation	300	4
3	AVE092M513	Creative Studio V (Compositing for 2D Animation)	300	4
4	AVE092M514	Motion Graphics	300	4
Minor Papers				
5	AVE092N511	Basics of Immersive Experience	300	4

Internship				
6	AVE092M524	Internship	300	4
Total Credit for 5th Semester				24

6th Semester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M611	Advance Visual Effects	300	4
2	AVE092M612	Foundation of Immersive Experience	300	4
3	AVE092M613	AI in Animation	300	4
4	AVE092M614	Creative Studio VI (Portfolio Building/Showreel)	300	4
Minor Papers				
5	AVE092N611	Basics of 2D Animation	300	4
Total Credit for 6th Semester				20
7th Semester				
Sl. No.	**Subject Code**	**Name of Subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M711	Introduction to Game Design	400	4
2	AVE092M712	Creative Research for Animation Production	400	4
3	AVE092M713	Visual Effects Production	400	4
4	AVE092M714	UI/UX for Interactive Media	400	4
Minor Papers				
5	AVE092N711	Introduction to Architecture Modeling	400	4
Total Credit for 7th Semester				20
Above 7.5 CGPA (Research + Honours)				
8th Semester				
Sl. No.	**Subject Code**	**Name of Subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M801	Media Entrepreneurship	400	4
Minor Papers				
2	AVE092N811	AI Generated Short Film	400	4
3	AVE092M812	**Project Dissertation / Research based Project** (Preferred Option)	400	12
Total Credit for 8th Semester				20
Below 7.5 CGPA (Honours)				
8th Semester				
Sl. No.	**Subject Code**	**Name of Subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M801	Media Entrepreneurship	400	4
Minor Papers				
2	AVE092N811	AI Generated Short Film	400	4
Major Core Papers				
3	AVE092M813	Major Project	400	12
Total Credit for 8th Semester				20

Semester I	
Major Paper-1	History of Animation and Multimedia
Subject Code	AVE092M101
Credit	3
L-T-P-C	2-1-0-3
Scheme of Evaluation	Theory

Course Objective: New interpretations of contemporary ideas of animation based on an understanding of the history of animation.

Course Outcomes:

On successful completion of the course, the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Find** the meaning, concept, and process of animation.	**BT 1**
CO 2	**Classify** the characteristics of the different types of animation.	**BT 2**
CO 3	**Develop** the principles of animation theories in Multimedia.	**BT 3**
CO 4	**Analyze** the composition associated with the rise and evolution of animation.	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**History of Animation** Influence of predecessors,1888–1909: Earliest animations on film, 1910s: From original artists to "assembly-line" production studios,1920s: Absolute film, synchronized sound and the rise of Disney,1930s: Color, depth, cartoon superstars,1950s: Shift from classic theatrical cartoons to limited animation in TV series for children,2000s–2010s: traditional techniques overshadowed by computer animation.	**15**
II	**History of Mean Methods of Animation** Magic Lanterns and the zoetrope, Puppet Animation, Animated Series, Movies, History of comic and manga, Celluloid Animation, 2D Animation,3D Animation, , Stop Motion, Motion Graphics	**15**
III	**History of Anime** Origins of Anime (early 1900-1922), Pre-war productions (1923-1939), During the Second World War, Post-war environment, Toei Animation and Mushi Production	**10**
IV	**Case Studies** Study about Animation film.	**5**
	Total	**45**

Text Books:

- Blaise, Aaron. (2021). *The Art of Aaron Blaise*. Vol.1. ISBN 1737328801.

- Eugene, Emile. &Courted, Jean Louis. (1908). *Fantasmagorie.* Movie

Reference Books:

- "Of Mice and Magic: A History of American Animated Cartoons**"** by *Leonard Maltin*
  (A detailed history of animation in America with insights into studios, techniques, and key figures.)

- "Animation: A World History" (3 Volumes) by Giacomo Miceli
  (A comprehensive series covering global animation history, styles, and evolution.)

Semester I	
Major Paper-2	Fundamentals of Drawing for Animation
Subject Code	AVE092M112
Credit	3
L-T-P-C	0-0-6-3
Scheme of Evaluation	Practical

Course Objective: To equip the students with the foundational concepts of art that will enable them tounderstand, draw different art styles, study human and animal anatomy.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Recall** the basics of skeleton, muscle, proportion of animal and human anatomy to portray the art in the most compelling way.	**BT 1**
CO 2	**Demonstrate** the understanding of shapes and forms, styles, and traditions through familiarization with a wide range of design, styles, etc.	**BT 2**
CO 3	**Apply** the knowledge of art to draw prospective drawing and human figures.	**BT 3**
CO 4	**Classify** new interpretations of contemporary ideas of design based on the understanding of Principles of Designs and Elements of Shape.	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**Art Fundamentals** Elements of Designs Shape, Line, shape, form, color, value, texture, and space, Doubt Clearing Assignments	**20**
II	**Human and Animal's Anatomy** Soft cube theory, Eye Pupils, Eye Shapes, Eye Expressions, Noses, Mouths, Mouth Shapes, Cheeks, Lips, Ears, Head Shapes, Facial Features, Hair, Body Variations (Male and Female), Pelvis and Legs, Simplified Anatomy Legs, Arms and Torso, Hand Feet Animal in General, Bird, Fish, Small Furry Creatures, Creepy Crawlies, Reptiles and Amphibians	**30**
III	**Creating Background Using Perspectives** Exterior, Interior, Angle Design by Applying Linear Perspective; Still life Drawing and Composition, Pencil, Black and White and Color, Rendering; Doubt Clearing Assignments	**20**
IV	**Creative Practical Application** Assignments	**20**
TOTAL		**90**

Text Books:

- Drawing and Anatomy by Victor Perard (1928)
- Illusion of Life by Frank Thomas and Olli Johnston (1981)

Reference Books:

- "Figure Drawing for All It's Worth" by Andrew Loomis
  (Essential for understanding anatomy and proportion, foundational for animators.)
- **"Force: Dynamic Life Drawing for Animators"** by *Michael D. Mattesi*
  (Focuses on gesture, energy, and motion, which are crucial in animation drawing.)

Semester I	
Minor Paper-1	Foundations of Colour Theory
Subject Code	AVE092N111
Credit	3
L-T-P-C	0-0-6-3
Scheme of Evaluation	Practical

Course Objective: The objective of Color Theory & Abstract is to enable the students to develop the knowledge of color and its applications in different scenarios.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Define** the application and uses of color	**BT 1**
CO 2	**Classify** the color terminologies and theories	**BT 2**
CO 3	**Develop** the different color schemes on compositions	**BT 3**
CO 4	**Analyze** color psychology in real world scenarios	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**Colour Wheel** Colour wheel – Primary, Secondary and Tertiary Colours, Warm and Cool colour	**10**
II	**Color Psychology** Color Psychology Visual Hierarchy, Colour Schemes, Colour Harmony	**15**
III	**Grey Scale** Grey scale method (Value and Colour)	**10**
IV	**Creative Practical Application** Assignments	**10**
	TOTAL	**45**

Textbooks:

- Interaction of Color by Josef Albers
- The Secret Lives of Color by Kassia St. Clair

Reference Books:

- The Color Bible: The definitive guide to color in art and design by Laura Perryman
- "Interaction of Color" by Josef Albers

  (A seminal work in color theory, perfect for deep understanding of color relationships.)

(A seminal work in color theory, perfect for deep understanding of color relationships.)

Semester I	
Skill Enhancement Courses -1	Creative Studio I (Characters & Life Studies)
Subject Code	AVE092S111
Credit	3
L-T-P-C	0-0-6-3
Scheme of Evaluation	Practical

Course Objective: A personal illustration style and technique that uses both traditional and digital skills and incorporates acquired knowledge, experience, judgement, and unique aesthetic vision.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Bloom's Taxonomy Level**
CO 1	**Define** a character's traits, personality, backstory according to the story	**BT 1**
CO 2	**Relate** different styles through which students can sketch different themes or character design props.	**BT 2**
CO 3	**Build** the different techniques in illustration.	**BT 3**
CO 4	**Analyze** themes of different conceptual art of animation	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**Art Fundamentals & Still life** Stick Figures, Gesture Study (Still Life), Props (Still life) Color Theory, Doubt Clearing Assignments	**30**
II	**Character forms and proportion** Introduction to Character Drawing with Simple, Mix and Complex Shape, Shape Modification, Expression Sheet, Construction Poses -Angles, Rhythm and Twist, Weight, Balance and Silhouettes, Doubt Clearing Assignments	**20**
III	**Hair Cloth Dynamics** Fundamental of drawing hair, Basic Components of drawing hair, drawing basic hair shapes, drawing clothing and cloth dynamics such as fold etc. Doubt Clearing Assignments	**20**
IV	**Developing Character from Imagination** Inspiration board. Character Style Study, Inking Process, Character Concept Art, Pose Board, Character Turn Around, Adding Character Personality, Character Props, Assignment: - Mock Client brief, Portfolio Building.	**20**
TOTAL		**90**

Textbooks:

- Creating Stylized Characters by Marisa Lewis
- The Silver Way: Techniques, Tips, and Tutorials by Stephen Silver

Reference Books:

- "Character Design from the Ground Up" by Kevin Crossley
  (Practical and creative advice on creating compelling characters with abstract or stylized features.)
- "Creating Characters with Personality" by Tom Bancroft (former Disney animator)
  (A practical guide that walks you through developing unique, animated characters from idea to final design. Includes interviews with top industry artists.)

Semester I	
VAC	Introduction to Speculative Design
Subject Code	VAC992V1409
Credit	3
L-T-P-C	0-0-3-3
Scheme of Evaluation	Practical

Course Objective: This course introduces students to the principles and methodologies of Design Thinking as a human-centered approach to problem-solving. Students will learn to identify user needs, define problems, generate innovative ideas, develop prototypes, and test solutions through iterative processes. The course emphasizes creativity, empathy, collaboration, critical thinking, and practical application of design methods to address real-world challenges.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Identify** future possibilities and societal impacts of emerging technology and shifting cultural and social trends, diverging from traditional design to provoke new thought, debate, directions, preferences, and strategy.	**BT 1**
CO 2	**Understand** user-centered research methods and problem-definition techniques.	**BT 2**
CO 3	**Apply** Scenario Building, Storytelling and Narratives, Interdisciplinary Approach to generate ideas, develop prototypes, and solve design problems.	**BT 3**
CO 4	**Analyze** user feedback and refine solutions through iterative design processes.	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**Understanding Speculative Design:** Future Studies of Design Aspects, Real World Case Study and Exploring Future Possibilities, Observation techniques, Ethical Considerations and Responsibility.	10
II	**Ideation & Concept Development:** Understanding the future trends, Digital and Manual Drawing, Brainstorming Techniques for Future Work Design Aspects, Mind Mapping, Idea Evaluation and selection.	10
III	**Methods and Approaches**:- Scenario Building, Storytelling and Narratives, Interdisciplinary Approach, Concept sketching, Storyboarding, Collaborative ideation work	10
IV	**Prototyping & Testing:** Refinement of Solutions, Final project	15
TOTAL		**45**

Textbooks

1. Brown, Tim. **Change by Design: How Design Thinking Creates New Alternatives for Business and Society**. Harper Business, 2009.
2. Ambrose, Gavin & Harris, Paul. **Design Thinking: Basics for Designers**. Bloomsbury Publishing, 2010.
3. Liedtka, Jeanne & Ogilvie, Tim. **Designing for Growth: A Design Thinking Tool Kit for Managers**. Columbia Business School Publishing, 2011.

Reference Books

1. Kelley, Tom & Kelley, David. **Creative Confidence: Unleashing the Creative Potential Within Us All**. Crown Business, 2013.
2. Norman, Don. **The Design of Everyday Things**. Revised and Expanded Edition, Basic Books, 2013.
3. Martin, Roger. **The Design of Business: Why Design Thinking is the Next Competitive Advantage**. Harvard Business Review Press, 2009.

Semester I	
Interdisciplinary Course (IDC-1)	Introduction to Indian Knowledge System - I
Subject Code	IKS992I101
Credit	3
L-T-P-C	2-1-0-3
Scheme of Evaluation	Theory (70%) + Continuous Evaluation (30%)

Credit Distribution (hours)		
L/T	P	EL
60	0	30

Course objective: This Foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG program. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Trad
itions over the centuries.

Course Outcomes:

On completion of this course students will be able to –

CO Level	Course Outcomes	BT Level
CO1	**Recall** the rich heritage of Indian knowledge systems	BT level 1
CO2	**Describe** the contribution of Indian knowledge systems to the world	BT level 2
CO3	**Demonstrate** knowledge of sociocultural and ethnolinguistic diversity that constitutes the soul of Bharatvarsha	BT level 2
CO4	**Apply** traditional knowledge and techniques in day-to-day life	BT level 3
CO5	**Distinguish** knowledge traditions that originated in the Indian subcontinent	BT level 3

Module	Course Contents	Periods
I	**Introduction to Indian Knowledge Systems (IKS):** -What is the Indian Knowledge System? -Definition of Indigenous/ Traditional Knowledge -Scope, and Importance of Traditional Knowledge. **Ancient India- Bharat Varsha:** -People of Ancient Bharat Varsha -Our great natural heritage: The great Himalayas and the rivers. - The civilizations of the Sindhu-Ganga valley, and the Brahmaputra valley. -Our coastal plains. -Our Nature: Forests and Minerals -Ancient Indian Traditional Knowledge and Wisdom about nature and climate.	15
II	**Indian Heritage of Knowledge:** -Ancient Indian Knowledge: The *Vedas* and its components-the *Vedangas* -Ancient Indian Books and treaties: The *Sastras*. -The Great Indian Epics: The Ramayana and The Mahabharata, -Epics and religious treaties of ancient Assam: Introduction to Madhav Kandali's *Ramayan* and Srimanta Sankardev's *Dasam Skandha Bhagavat* of the Puranas. -Ancient Traditional Knowledge-The *Agamas* -The ancient Buddhist knowledge: *Tripitaka: Vinaya, Sutta* and *Abhidhamma Pitaka* **Languages and language studies in India**: -What is linguistics? -Script and Language -Alphabet of the Indian languages *Varnamala:* Origin, Evolution, and phonetic features. -Languages of India -Important texts of Indian languages: Skills *Siksha,* Expression/Prounciation-*Nirukta,* Grammer-*Vyakarana,* Poetic rhythm- *Chandas.* -Paninian Grammar: A Brief Introduction **Introduction to Fine Arts and Performing Arts of India:** -Ancient Indian classical music and dance forms: The Science of Dramas-*Natyasastra* and the Science of Music-*Gandharva-Veda.* -Aesthetics in Indian Art and Culture. -Folk music and traditional dance forms of the Northeast.	15
III	**Indian Science & Technology** -Ancient India's contribution to Mathematics- Number System. Algebra and Arithmetic, Geometry and Trigonometry. -Origin of Decimal system in India; nomenclature of numbers in the Vedas. Zero and Infinity. Sulba-sutras. Contribution of Brahmagupta and Sridhar Acharya to Mathematics. Important texts of Indian mathematics. • **Indian Astronomy:** Planetary System. Motion of the Planets. Velocity of Light. Eclipse. Astronomy. Navagrahas. Important works in Indian Astronomy. Aryabhata and Nilakantha: Contribution to Astronomical Studies • **Indian Metal Works:** Mining Techniques. Types of Metals. Tools & Techniques for Metal Smelting with examples. Metalworks in pre-modern India: Special reference to NE India.	15

IV	**Contribution of Ancient India to Health Sciences:** -Traditional Indigenous systems of medicines in India: *Ayurveda* and *Yoga:* Elements of *Ayurveda: Gunas* and *Doshas*, *Pancha Mahabhuta* and *Sapta-dhatu*. -Concept of disease in Ayurveda -Ayurvedic lifestyle practices: *Dinacharya* and *Ritucharya.* -Important Ayurvedic Texts -Hospitals in Ancient India -*Ayurveda*: Gift of India to the modern world.	15
EL	The experiential learning sessions may include: • Field Visits: Organizing visits to historical sites, museums, traditional craft centers, and other places relevant to Indian knowledge systems. • Interactive Sessions: Engaging students in discussions with experts and practitioners in various fields of Indian knowledge systems to gain insights and practical knowledge. • Online Lecture Series: Providing the students with online lectures by distinguished experts in the field of the Indian Knowledge System. • Hands-on Activities: Providing opportunities for students to participate in activities related to traditional arts, crafts, music, dance, agriculture, etc., to understand the practical aspects of Indian knowledge systems. Practical Demonstrations: Conducting workshops or sessions to demonstrate traditional practices, such as yoga, Ayurveda, Vastu Shastra, etc., for the students.	30
	Total	90

Textbooks:

- Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022), *Introduction to Indian Knowledge System: Concepts and Applications.* PHI Learning Private

Semester I	
AEC-1	CEN I: Introduction to Effective Communication
Subject Code	CEN982A101
Credit	2
L-T-P-C	2-0-0-2
Scheme of Evaluation	Theory and Practical

Course Objective: To understand the four major aspects of communication by closely examining the processes and outlining the most effective ways to communicate with interactive activities.

Course Outcomes: On successful completion of the course the students will be able to

CO Level	Course Outcomes	Blooms Taxonomy Level
CO 1	**List** the elements and processes that make for successful communication and **recognize** everyday activities that deserve closer attention to improve communication skills	**BT 1**
CO 2	**Contrast** situations that create barriers to effective communication and relate them to methods that are consciously devised to overcome such hindrance	**BT 2**
CO 3	**Apply** language, gestures, and paralanguage effectively to avoid miscommunication and articulate one's thoughts and build arguments more effectively	**BT 3**

Course Outlines:

Course Contents		
Units	**Course Contents**	**Periods**
I	Introduction to Effective Communication • Listening Skills o The Art of Listening o Factors that affect Listening o Characteristics of Effective Listening o Guidelines for improving Listening skills	**5**
II	• Speaking Skills o The Art of Speaking o Styles of Speaking o Guidelines for improving Speaking skills o Oral Communication: importance, guidelines, and barriers	**5**

III	• Reading Skills ○ The Art of Reading ○ Styles of Reading: skimming, surveying, scanning ○ Guidelines for developing Reading skills	**5**
IV	• Writing Skills ○ The Art of Writing ○ Purpose and Clarity in Writing ○ Principles of Effective Writing	**5**

Textbooks:

1. Rizvi, M. Ashraf. (2017). *Effective Technical Communication*. McGraw-Hill.
2. Chaturvedi, P. D. and Chaturvedi, Mukesh. (2014). *Business Communication.* Pearson.
3. Raman, Meenakshi and Sharma, Sangeeta. (2011). *Technical Communication: Principles and Practice* (2nd Edition): Oxford University Press.

Semester III	
Major Course - 1	Tradigital Animation
Subject Code	AVE092M311
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course covers the full Tradigital Animation production process, from planning to final output. Students will learn to create model sheets, storyboards, animatics, and develop skills in keyframing, in-betweening, lip-syncing and coloring.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Relate** the storyboard and animatic *sequence* and overall flow of the story and whether the images effectively convey the intended narrative.	**BT 1**
CO 2	**Compare** each movement timing, spacing, and squash and stretch to create a sense of weight, momentum, and fluidity in the animation	**BT 2**
CO 3	**Organize** with technical proficiency, and attention to detail to create smooth and realistic 2D animation that captures the essence of the movement trying to depict.	**BT 3**
CO 4	**Analyze** the knowledge of Tradigital animation	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Industrial Production Pipeline:** Plan animation production and set deadline for Tradigital Animation production process	**30**
II	**Visual World Building:-** Concept Art Design, Characters, Background and Prop, Layout, Scratch Sound, Animatic	**30**
III	**Animation Project Production:-** Keyframe animation, in-between animation, test, clean up, test, lip-sync, Inking and Coloring	**30**
IV	**Project:** Students will have to submit a project	**30**
	TOTAL	**120**

Textbooks:

- Williams, Richard. (2001). *The Animator's Survival Kit*
- *Timing for Animation, 40th Anniversary Edition (greyscale)*

Reference Books:

- "Animation 1: Learn to Animate Cartoons Step by Step" by author: Preston Blair
- "The Illusion of Life: Disney Animation" by author *Frank Thomas & Ollie Johnsto*

Semester III	
Major Course - 2	Introduction to 3D Modelling
Subject Code	AVE092M312
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course introduces students to the fundamental concepts and techniques of 3D modeling used in animation, gaming, product visualization, and digital media production. Students will learn the 3D modeling pipeline, interface navigation, polygonal modeling techniques, object creation, topology, and scene organization. The course emphasizes developing accurate and efficient 3D models while understanding form, proportion, and modeling workflows used in industry-standard software.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Identify** the fundamentals of 3D modeling, modeling workflows, and the 3D production pipeline.	**BT 1**
CO 2	**Understand** polygonal modeling techniques, topology, and object construction principles.	**BT 2**
CO 3	**Apply** modeling tools and techniques to create 3D objects and environments.	**BT 3**
CO 4	**Analyze** and refine 3D models by optimizing topology, proportions, and scene organization.	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**Introduction to 3D Modeling & Interface:** Overview of 3D graphics and modeling pipeline, Introduction to 3D software interface, Navigation tools & viewports, Object transformation (Move, Rotate, Scale), Primitive objects, Scene hierarchy, File management and project organization.	**30**
II	**Polygonal Modeling Fundamentals:** Polygon components (Vertex, Edge, Face), Extrude, Inset, Bevel, Loop cuts, Modeling simple objects from primitives, Object editing techniques, Understanding topology, Low-poly modeling principles.	**30**
III	**Advanced Object Modeling:** Hard-surface modeling techniques, Creating household and product models, Edge flow and topology refinement, Symmetry and mirror workflows, Modifier-based modeling, Model cleanup and optimization, NURBS Modelling.	**30**
IV	**Environment & Scene Development:** Creating simple environmental assets, Scene assembly and organization, Basic materials assignment, Model optimization, Preparing assets for rendering and animation, Final modeling project and presentation.	**30**
TOTAL		**120**

Textbooks:

- Anatomy for Sculptors by U. Zarins and S. Kondrats
- **Autodesk Maya 2026 Basics Guide** by Kelly L. Murdock.

Reference Books:

- "**Pushing Points Topology Workbook:** By William Vaughan.
- "The Illusion of Life: Disney Animation" by author *Frank Thomas & Ollie Johnsto*

Semester III	
Minor Course -I	Stopmotion and Cutout Animation
Subject Code	AVE092N311
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: Student shall be guided on to execute Stop Motion Animation. Student shall cover every step in details, ranging from what and how a story for stop motion animation should be and how does it differ from other forms of storytelling

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms TaxonomyLevel**
CO 1	**Understand** interface and workflow of the software for effective work	**BT 1**
CO 2	**Review** the storyboard overall flow of the story and whether the images effectively convey the intended narrative.	**BT 2**
CO 3	**Apply** the knowledge of color, shapes and forms to retain mood and the mass of subject to create smooth *sequence.*	**BT 3**
CO 4	**Demonstrate** the knowledge of stop motion	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction** What is Stop motion, Type of Stop motion, Animation Principles Interface of software, Interface, Tool, Timeline, Properties, Introduction to setup and shooting	**15**
II	**Storyboard and Staging** Script, Storyboard, Character and Prop Building, colour schemes and psychology, Stage setup,	**15**
III	**Cinematic Light and Shoot** Type of Lights, Setting the Mood and Tone, Film Lighting Techniques, Camera and Lens, Shooting	**15**
IV	**Creative Practical Application** Assignments: - Video Reel	**15**
	TOTAL	**60**

Textbooks:

- Stop Motion Filmmaking: The Complete Guide to Fabrication and Animation by Christopher Walsh
- Stop Motion: Craft Skills for Model Animation by Susannah Shaw

Reference Books:

- The History and Theory of Cut-Out Animation (Dan Torre)
- The Art of Stop-Motion Animation (Ken A. Priebe)

Semester III	
SEC - III	Creative Studio III (Design Thinking Process)
Subject Code	AVE092S311
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course introduces students to the principles and methodologies of Design Thinking as a human-centered approach to problem-solving. Students will learn to identify user needs, define problems, generate innovative ideas, develop prototypes, and test solutions through iterative processes. The course emphasizes creativity, empathy, collaboration, critical thinking, and practical application of design methods to address real-world challenges.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Identify** the principles, stages, and importance of the Design Thinking process.	**BT 1**
CO 2	**Understand** user-centered research methods and problem-definition techniques.	**BT 2**
CO 3	**Apply** Design Thinking tools to generate ideas, develop prototypes, and solve design problems.	**BT 3**
CO 4	**Analyze** user feedback and refine solutions through iterative design processes.	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**Introduction to Design Thinking:** Overview of Design Thinking, Human-centered design principles, Understanding innovation and creativity, Design Thinking process and its applications, Introduction to problem-solving frameworks, Case studies of successful design solutions.	30
II	**Empathize & Define:** User research methods, Observation techniques, Interviews and surveys, Empathy mapping, User personas, Customer journey mapping, Identifying user needs and pain points, Problem framing, How Might We (HMW) statements.	30
III	**Ideation & Concept Development:** Brainstorming techniques, Mind mapping, SCAMPER method, Lateral thinking, Idea evaluation and selection, Concept sketching, Storyboarding, Collaborative ideation workshops.	30
IV	**Prototyping & Testing:** Low-fidelity and high-fidelity prototyping, Rapid prototyping techniques, User testing methods, Feedback collection and analysis, Iterative design process, Refinement of solutions, Final Design Thinking project presentation.	30
TOTAL		**120**

Textbooks

4. Brown, Tim. **Change by Design: How Design Thinking Creates New Alternatives for Business and Society**. Harper Business, 2009.
5. Ambrose, Gavin & Harris, Paul. **Design Thinking: Basics for Designers**. Bloomsbury Publishing, 2010.
6. Liedtka, Jeanne & Ogilvie, Tim. **Designing for Growth: A Design Thinking Tool Kit for Managers**. Columbia Business School Publishing, 2011.

Reference Books

4. Kelley, Tom & Kelley, David. **Creative Confidence: Unleashing the Creative Potential Within Us All**. Crown Business, 2013.
5. Norman, Don. **The Design of Everyday Things**. Revised and Expanded Edition, Basic Books, 2013.
6. Martin, Roger. **The Design of Business: Why Design Thinking is the Next Competitive Advantage**. Harvard Business Review Press, 2009.

Semester III	
Interdisciplinary Course	Traditional Art and Contemporary Design Integration
Subject Code	AVE092I301
Credit	3
L-T-P-C	3-0-0-3
Scheme of Evaluation	Theory

Course Objective: This course explores the influence of traditional Indian art forms on contemporary animation design. Students will study the visual language, symbolism, motifs, storytelling traditions, and aesthetic principles of folk and tribal arts and examine their application in modern animation practices. The course emphasizes the adaptation of traditional artistic elements into character design, visual development, narrative construction, motion design, and digital storytelling, fostering culturally rooted yet contemporary animation concepts.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Identify** and explain the visual characteristics, cultural significance, and storytelling traditions of selected traditional Indian art forms.	**BT 1**
CO 2	**Interpret** traditional motifs, symbols, narratives, and aesthetic principles for animation and visual storytelling applications.	**BT 2**
CO 3	**Apply** traditional artistic elements and visual languages in the conceptual development of animation characters, environments, and narratives.	**BT 3**
CO 4	**Analyze** and evaluate the integration of traditional art forms into contemporary animation design while maintaining cultural relevance and authenticity.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction to Traditional Indian Art Forms:** Overview of folk and tribal arts; Madhubani, Warli, Pattachitra, Gond, Kalamkari and other regional traditions; cultural contexts; visual language; motifs, symbols, patterns, color systems, and narrative traditions in traditional art.	**10**
II	**Visual Storytelling and Design Translation:** Analysis of traditional narratives, mythology, folklore, and symbolic representation; abstraction and stylization techniques; adaptation of traditional visual elements into animation-friendly forms; principles of visual storytelling.	**10**
III	**Traditional Art in Animation Design:** Application of folk and tribal art aesthetics in character design, background design, prop design, visual development, storyboarding, title sequences, and animated	**10**

	narratives; case studies of traditional art-inspired animation projects.	
IV	**Contemporary Digital Applications:** Integration of traditional art aesthetics into motion graphics, digital storytelling, interactive media, animated educational content, game art, and immersive experiences; cultural preservation through animation and emerging media.	**15**
	TOTAL	**45**

Textbooks:

1. *Madhubani Art: Indian Art Series* – Bharti Dayal
2. *Indian Folk and Tribal Paintings* – B. N. Goswamy
3. *Traditional Indian Painting* – Ananda K. Coomaraswamy
4. *Animation: From Script to Screen* – Shamus Culhane

Reference Books:

1. *Indian Art and Culture* – Nitin Singhania
2. *The Arts of India* – Susan L. Huntington & John C. Huntington
3. *The Illusion of Life: Disney Animation* – Frank Thomas and Ollie Johnston
4. *Directing the Story* – Francis Glebas
5. *Visual Development for Animation* – Relevant contemporary animation design references

Semester III	
AEC-3	CEN III – Fundamentals of Business Communication
Subject Code	CEN982A301
Credit	2
L-T-P-C	2-0-0-2
Scheme of Evaluation	Theory and Practical

Course Objective: The aim of the course is to develop essential business communication skills, including effective writing, speaking, and interpersonal communication, to enhance professional interactions, collaboration, and successful communication strategies within diverse corporate environments.

Course Outcomes: On successful completion of the course the students will be able to:

CO Level	Course Outcomes	Blooms Taxonomy Level
CO 1	Define and list business documents using appropriate formats and styles, demonstrating proficiency in written communication for various business contexts.	**BT 1**
CO 2	Demonstrate confident verbal communication skills through persuasive presentations, active listening, and clear articulation to engage and influence diverse stakeholders.	**BT 2**
CO 3	Apply effective interpersonal communication strategies, including conflict resolution and active teamwork, to foster positive relationships and contribute to successful organizational communication dynamics	**BT 3**

Course Outlines:

Units	Course Contents	Periods
I	**Business Communication: Spoken and Written** • The Role of Business Communication • Classification and Purpose of Business Communication • The Importance of Communication in Management • Communication Training for Managers • Communication Structures in Organizations • Information to be Communicated at the Workplace • Writing Business Letters, Notice, Agenda and Minutes	5
II	**Negotiation Skills in Business Communication** • The Nature and Need for Negotiation o Situations requiring and not requiring negotiations • Factors Affecting Negotiation o Location, Timing, Subjective Factors • Stages in the Negotiation Process o Preparation, Negotiation, Implementation • Negotiation Strategies	5
III	**Ethics in Business Communication** • Ethical Communication • Values, Ethics and Communication • Ethical Dilemmas Facing Managers • A Strategic Approach to Business Ethics • Ethical Communication on the Internet • Ethics in Advertising	5
IV	**Business Etiquettes and Professionalism** • Introduction to Business Etiquette • Interview Etiquette • Social Etiquette • Workplace Etiquette • Netiquette	5
	Total	20

Textbooks:

1. Business Communication *by Shalini Verma*
2. Business Communication *by P.D. Chaturvedi and Mukesh Chaturvedi*

Semester III	
AEC-2	Behavioural Science -III
Subject Code	BHS982A302
Credit	1
L-T-P-C	1-0-0-1
Scheme of Evaluation	Theory

Course objective: To increase one's ability to draw conclusions and develop inferences about attitudes and behaviour, when confronted with different situations that are common in modern organizations.
To enable the students to understand the process of problem solving and creative thinking.

Course outcomes: On completion of the course the students will be able to:

CO1: Understand the process of problem solving and creative thinking.
CO2: Develop and enhance of skills required for decision-making.

Modules	Course Contents	Periods
I	**Problem Solving Process** Defining problem, the process of problem solving, Barriers to problem solving (Perception, Expression, Emotions, Intellect, surrounding environment)	4
II	**Thinking as a tool for Problem Solving** What is Thinking: The Mind/Brain/Behaviour Critical Thinking and Learning: -Making Predictions and Reasoning. -Memory and Critical Thinking. - Emotions and Critical Thinking.	4
III	**Creative Thinking** - Definition and meaning of creativity, - The nature of creative thinking: Convergent and Divergent thinking, - Idea generation and evaluation (Brainstorming) - Image generation and evaluation. - The six-phase model of Creative Thinking: ICEDIP model	4
IV	**Building Emotional Competence** Emotional Intelligence – Meaning, components, Importance and Relevance Positive and Negative emotions Healthy and Unhealthy expression of emotions	4
Total		**16**

Textbooks:

- J William Pfeiffer (ed.) Theories and Models in Applied Behavioural Science, Vol 3, Management; Pfeiffer &Company
- Blair J. Kolasa, Introduction to Behavioural Science for Business, John Wiley & Sons Inc.

Semester V	
Major Course:1	3D Animation Techniques
Subject Code	AVE092M511
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course introduces students to the principles and techniques of three-dimensional animation. Students will learn the fundamentals of motion, timing, spacing, character performance, and animation workflows. The course emphasizes applying classical animation principles to 3D environments, enabling students to create believable movements and engaging animated sequences for digital media, entertainment, and interactive experiences.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Identify** the principles of animation and the stages involved in a 3D animation production pipeline.	**BT 1**
CO 2	**Understand** concepts of motion, timing, spacing, weight, and character performance in animation.	**BT 2**
CO 3	**Apply** animation principles to create object, camera, and character animations.	**BT 3**
CO 4	**Analyze** and refine animated sequences to improve realism, appeal, and storytelling effectiveness.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
1	**Fundamentals of Animation:** History and evolution of animation, Applications of Principles of animation, Timing and spacing, Keyframes and interpolation, Motion analysis, Weight and balance, Observation and movement studies, Introduction to animation workflow.	30
2	**Object and Camera Animation:** Animating simple objects, Translational and rotational motion, Bounce, pendulum, overlap and follow-through, Camera movements, Camera composition, Creating dynamic shots, Animation curves and motion refinement.	30
3	**Character Animation Principles:** Walk cycles, Run cycles, Body mechanics, Acting and performance, Facial expressions, Lip-sync fundamentals, Secondary motion, Character posing, Conveying emotions through movement.	30
4	**Animation Production and Final Project:** Storytelling through animation, Shot planning, Animation polishing techniques, Refinement of timing and spacing, Animation review process, Rendering animated sequences, Final animation project and presentation.	30
	TOTAL	**120**

Textbooks

1. Beane, Andy. **3D Animation Essentials**. Sybex, 2nd Edition, 2012.
2. 3D Animation Essentials by Andy Beane:

Reference Books

1. Johnston, Ollie & Thomas, Frank. **The Illusion of Life: Disney Animation**. Disney Editions, 1995.
2. Kerlow, Isaac Victor. **The Art of 3D Computer Animation and Effects**. Wiley, 4th Edition, 2009.
3. Halas, John & Whitaker, Harold. **Timing for Animation**. Focal Press, 2009.

Semester V	
Major Course:2	Lighting for Animation
Subject Code	AVE092M512
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course introduces students to the artistic and technical principles of lighting in animated productions. Students will learn how lighting contributes to storytelling, mood, atmosphere, visual continuity, and character emphasis. The course focuses on cinematic lighting techniques, color psychology, scene composition, and lighting workflows used in animation production. Students will develop the skills necessary to create visually compelling animated sequences through effective lighting design.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Identify** the role of lighting in animation, visual storytelling, and cinematic production.	**BT 1**
CO 2	**Understand** lighting principles, color theory, mood creation, and scene composition for animated environments.	**BT 2**
CO 3	**Apply** lighting techniques to enhance character performance, environments, and narrative communication.	**BT 3**
CO 4	**Analyze** and refine lighting setups to achieve visual consistency, dramatic impact, and storytelling effectiveness.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
1	**Introduction to Lighting:** Understanding light and its importance in animation, Natural and artificial light sources, Properties of light, Light direction, Intensity and color, Observation of light in everyday environments, Introduction to digital lighting concepts.	30
2	**Basic Lighting Techniques:** Types of lights used in 3D scenes, Creating simple lighting setups, Three-point lighting, Front, side and back lighting, Shadows and their importance, Lighting simple objects and scenes.	30
3	**Lighting for Mood and Visual Appeal:** Understanding color and mood, Warm and cool lighting, Day and night lighting, Indoor and outdoor lighting, Creating atmosphere through lighting, Highlighting objects and characters using light.	30
4	**Scene Lighting and Presentation:** Lighting simple interior and exterior scenes, Balancing light and shadow, Introduction to rendering concepts, Preparing scenes for final output, Basic troubleshooting, Presentation of a simple animated scene with lighting.	30
	TOTAL	**120**

Textbooks

1. **Birn, Jeremy.** ***Digital Lighting and Rendering*****.** 4th Edition, New Riders Publishing, 2013.
    - Considered the standard textbook for understanding lighting principles, shadows, color, mood, and rendering workflows in computer graphics.
2. **Block, Bruce.** ***The Visual Story: Creating the Visual Structure of Film, TV and Digital Media*****.** 3rd Edition, Routledge, 2020.
    - Excellent for understanding how light, color, space, and visual composition contribute to storytelling.

Reference Books

1. **Alton, John.** ***Painting with Light*****.** University of California Press, 2013.
    - A classic text on lighting aesthetics and the creative use of light for visual storytelling.
2. **Brown, Blain.** ***Cinematography: Theory and Practice*****.** 4th Edition, Routledge, 2021.
    - Useful for understanding lighting design, mood creation, and visual communication from a cinematographic perspective.

Semester V	
Major Course:3	Creative Studio V (Compositing for 2D Animation)
Subject Code	AVE092M513
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course focuses on advanced 2D animation and visual composition techniques used in contemporary animation and motion graphics production. Students will develop skills in character performance animation, visual storytelling, parallax animation, camera movement, digital lighting, and compositing using industry-standard software. Through practical projects, learners will create visually engaging animated scenes that integrate animation, motion design, and visual effects workflows.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	Create rig-based character animations demonstrating acting, movement, and performance.	**BT 1**
CO 2	Design and animate scenes using advanced visual composition principles.	**BT 2**
CO 3	Develop multi-layered environments and parallax animations using After Effects.	**BT 3**
CO 4	Apply camera movements and cinematic techniques to enhance visual storytelling.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Advanced Character Animation and Performance** Review of Rig-Based Animation Workflow, Character Posing and Silhouette Design, Weight, Balance, and Body Mechanics, Acting Principles in Animation, Facial Expressions and Emotion, Dialogue and Lip-Sync Animation, Character Interaction and Performance, Scene Blocking and Timing Refinement Projects • Character Acting Shot • Lip-Sync Animation • Performance-Based Animation Sequence	**30**
II	**Visual Storytelling and Motion Design** Principles of Visual Composition, Storytelling Through Composition, Layout Design and Scene Planning, Camera Angles and Shot Composition, Motion Design Principles, Typography Animation, Motion Graphics Integration, Creating Dynamic Visual Narratives Projects • Animated Title Sequence • Motion Graphics Composition	**30**

	• Story-Based Animated Scene	
III	**Parallax Animation and Camera Techniques in After Effects** Introduction to 2.5D Animation Workflow, Scene Preparation for Parallax Animation, Layer Separation and Asset Organization, Creating Depth Using Multiple Layers, Working with 3D Layers in After Effects, Virtual Camera Setup, Camera Movement Techniques, Parallax Animation for Landscapes and Environments, Cinematic Scene Creation, Advanced Scene Transitions. Projects • Multi-Layer Environment Setup • Parallax Landscape Animation • Camera Fly-Through Sequence • Cinematic Environment Animation	**30**
IV	**Lighting, Compositing and Visual Effects in After Effects** Digital Lighting Fundamentals, Creating and Controlling Light Sources, Shadow Creation and Enhancement, Color Correction and Color Grading, Light Wrap and Depth Enhancement, Atmospheric Effects (Fog, Rain, Dust, Smoke), Particle Systems and Environmental Effects, Blending Modes and Adjustment Layers, Advanced Compositing Techniques, Rendering and Output Optimization Projects • Lighting and Mood Creation • Day-to-Night Scene Conversion • Atmospheric Effects Composition	**30**
	Total	**120**

Textbooks:

- Framed Ink: Drawing and Composition for Visual Storytellers by Marcos Mateu-Mestre
- The Visual Story: Creating the Visual Structure of Film, TV and Digital Media by Bruce Block

Reference books

- Professional Compositing and Visual Effects *by Ron Ganbar*
- Compositing Visual Effects: Essentials for the Aspiring Artist *by Steve Wright*

Semester V	
Major Course : 4	Motion Graphics
Subject Code	AVE092M514
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: The Motion Graphics course introduces students to the fundamentals of visual storytelling using Adobe Photoshop and Adobe After Effects. It focuses on developing skills in creating dynamic graphics, animated text, and simple transitions by combining design principles with animation techniques. Students will learn to design assets in Adobe Photoshop and animate them in Adobe After Effects, gaining a foundational understanding of keyframes, motion paths, timing, and compositing. The course aims to build creative confidence and technical proficiency for entry-level motion graphics work.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Recall** the basic tools, functions, and terminology used in Adobe Photoshop and Adobe After Effects.	**BT 1**
CO 2	**Explain** the principles of motion design and how design elements work in animated compositions	**BT 2**
CO 3	**Apply** the keyframes, transitions, and basic animation techniques to create motion graphics using Adobe After Effects.	**BT 3**
CO 4	**Analyze** the design elements and animation timing to enhance visual appeal and storytelling in motion graphics.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction to Adobe After Effects** Introduction to the UI of software, composition, tools, encoder, and rendering, Keyframe Animation	**15**
II	**Introduction to Motion Graphics** Creating holographic HUD, Motion posters, animating texts, shapes and masking, trim path, and different type of effects, Path Animation, Puppet Tool,	**15**
III	**Introduction to Effects** Parallax animation, understanding assets and effects, Chroma key, Tracking and Stabilizing and Mocha Tracking	**15**
IV	**Project** Students will have to individually submit projects of each sub contents in a storage device. Teacher will supervise the projects	**15**
	TOTAL	**60**

Textbooks:

- Motion Graphic Design: Applied History and Aesthetics *by Krasner*
- Design for Motion: Fundamentals and Techniques of Motion Design *by Austin Shaw*

Reference Books:

- The Motion Graphics Handbook *by McKissick*
- Creating Motion Graphics with After Effects *by Chris & Trish Meyer*

Semester V	
Minor 1	Basics of Immersive Experience
Subject Code	AVE092N511
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course introduces students to the fundamentals of interactive tools and technologies used for creating immersive digital experiences. The course focuses on interactive media principles, augmented reality concepts, spatial interaction, digital content integration, and user engagement. Students will explore the process of developing interactive experiences by combining visual assets, animation, audio, and real-world environments.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	Identify the fundamentals of interactive media and augmented reality technologies.	**BT 1**
CO 2	Understand interactive design principles, user engagement, and spatial interaction concepts.	**BT 2**
CO 3	**Apply** interactive tools and workflows to create basic augmented reality experiences.	**BT 3**
CO 4	**Analyze** and evaluate interactive experiences based on usability, engagement, and functionality.	**BT 4**

Course Outlines:

Modules	Course Contents	Periods
I	**Introduction to Interactive Media and AR:** Evolution of interactive technologies, Interactive media concepts, Introduction to Augmented Reality, Types of AR experiences, Applications of AR in education, entertainment, advertising, healthcare, retail, and cultural heritage, Components of interactive systems.	**15**
II	**Interactive Design Fundamentals:** Principles of interaction design, User-centered design, User engagement and experience, Information flow, Storyboarding for interactive experiences, Spatial interaction concepts, Designing interactions for real-world environments.	**15**
III	**AR Content Development:** Digital asset preparation, Integration of 2D and 3D content, Image-based tracking concepts, Marker-based and markerless interactions, Animation and audio integration, Interactive triggers and events, Creating simple interactive experiences.	**15**
IV	**Testing and Experience Design:** User testing methods, Performance optimization, Usability evaluation, Interaction refinement, Designing immersive user experiences, Documentation and presentation of interactive projects, Development of a basic AR experience as a final project.	**15**
		60

Textbooks

1. Craig, Alan B. **Understanding Augmented Reality: Concepts and Applications**. Morgan Kaufmann, 2013.
2. Schmalstieg, Dieter & Hollerer, Tobias. **Augmented Reality: Principles and Practice**. Addison-Wesley, 2016.
3. Carmigniani, Julie & Furht, Borko. **Handbook of Augmented Reality**. Springer, 2011.

Reference Books

1. LaViola, Joseph J., Kruijff, Ernst, McMahan, Ryan, Bowman, Doug & Poupyrev, Ivan. **3D User Interfaces: Theory and Practice**. Addison-Wesley, 2017.
2. Sherman, William R. & Craig, Alan B. **Understanding Virtual Reality: Interface, Application, and Design**. Morgan Kaufmann, 2018.
3. Jerald, Jason. **The VR Book: Human-Centered Design for Virtual Reality**. Association for Computing Machinery, 2015.

Semester VII	
Major Course 1	Introduction to Game Design
Subject Code	AVE092M711
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: Students construct their own games with visual storytelling, game mechanics and strategic gameplay. Devise rules that engage players and prototype components like cards, tokens and boards. Through hands-on playtesting and critique, refine your ideas into playable games that challenge and entertain.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Define** the basics of Game Play, Game theory, Equilibrium	**BT 1**
CO 2	**Interpret** the interface of the software and explore Nash Equilibrium	**BT 2**
CO 3	**Apply** the knowledge of Game Design to build the blueprint	**BT 3**
CO 4	**Analyze** the game prototype	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction to Game:** Game players, strategy, payoff, information set, equilibrium	**30**
II	**Game Theory Strategies and Equilibrium:** Types of game theory, Strategies, Nash Equilibrium, Software uses	**30**
III	**Game Design:** Types, game genres, GDD, Game environment, game component, Game Designer perspective, Player perspective, MDA frame work	**30**
IV	**Project:** Prototyping of Analog and Digital game concept	**30**
	TOTAL	**120**

Textbooks:

- The Books of Board Games: An Introduction to Modern Tabletop Gaming *by Oliver Roeder*
- Make Your Own Board Game: Designing, Building, and Playing an Original Tabletop Game *by Jesse Terrance Daniels*

Reference Books:

- Introduction to (Tabletop) Role-playing Game Design *by Brent Newhall*

Semester VII	
Major Course 2	Creative Research for Animation Production
Subject Code	AVE092M712
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Theory

Course Objective:

To make the students aware of research culture among academics and professionals in different fields. To undertake research in their specific academic fields on philosophical, epistemological understanding of the elements of research. To provide knowledge skills of various types of research designs and its procedures to conduct research to meet the national and international requirements.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Define** the different aspects of research.	**BT 1**
CO 2	**Demonstrate** the research methods critically, creatively, and independently	**BT 2**
CO 3	**Apply** the different research articles from different sources and categorize them	**BT 3**
CO 4	**Analyse** the evaluates in-depth information from diverse sources	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction to Research:** Definition & Concept (Variables, Hypothesis, Theory), Objectives, scope and importance of Research, Types of research, Approaches to research (quantitative and qualitative and mixed method); Steps in research, Concept of reliability, Validity.	**30**
II	**Methods & methodologies of Research:** Qualitative- Quantitative Technique, Content Analysis, Sample Survey Method, Observation Methods, Experimental Studies, Case Studies, Interview Method, Focus Groups, Ethnography Narrative Analysis, Historical research, Sampling techniques.	**30**
III	**Data Collection:** Data Collection Techniques, Primary Data, Secondary Data, Methods of collecting data, Statistical Analysis (Descriptive & Inferential Statistics), Central Tendency, Dispersion, Presentation, and Interpretation of Research Findings	**30**

IV	**Report Writing:** Academic Writing, Presentation of research reports, Referencing Management tool, Referencing and Citation Style, Ethics in Research	**30**
	TOTAL	**120**

Textbooks:

- *Mass Media Research*, Roger, Wimmer. D and Dominick, Joseph,R; Thomson Wadsworth; 2006.

Reference Books:

- Berger, Arthur Asa; *Media Research Techniques*; Second Edition; Sage Publications, New Delhi; 1998.
- Fiske, John; *Introduction to Communication Studies*; Third Edition; Routledge Publications; 1982.
- Croteau, David and Hoynes; *Media/Society: Industries, Images and Audiences;* William; Forge Press; 2002.
- Kothari, C.R; *Research Methodology: Methods and Techniques*; New Age International Ltd. Publishers; 2013.

Semester VII	
Major Course 3	Visual Effects Production
Subject Code	AVE092M713
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course offers a comprehensive overview of the VFX production pipeline, covering pre-production, 3D modeling, animation, FX simulations, and compositing. Students will learn to integrate CG with live-action footage and develop a professional showreel, preparing them for entry-level roles in film, TV, advertising, and gaming industries.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the VFX production pipeline and demonstrate an understanding of pre-production processes such as storyboarding, cinematography, and previs.	**BT 1**
CO 2	**Classify** 3D asset creation techniques, including modelling, texturing, shading, lighting, animation, and FX simulations, using industry-standard tools.	**BT 2**
CO 3	**Apply** compositing workflows, including match moving, rotoscoping, multi-pass compositing, colour grading, and integrating CG into live-action footage.	**BT 3**
CO 4	**Analyse** a final VFX project, demonstrating project planning, asset management, rendering optimization, and development of a professional showreel tailored for the VFX industry.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction to Visual Effects & Pre-Production** Understanding VFX & Industry Overview ,VFX Production Pipeline, Concept Development & Storyboarding, Introduction to Camera & Cinematography for VFX, Introduction to Previsualization (Previs)	**30**
II	**3D Asset Creation & Animation** 3D Modeling for VFX (3D Assets), Shading, Lighting & Rendering, Particle Effects & Simulations, Using Houdini / Blender for FX Simulations, Fire, Water, Smoke, Destruction FX, Procedural Workflows	**30**
III	**Compositing & post-production** Introduction to Compositing, Match moving & Rotoscoping, Advanced Compositing Techniques, 3D Projection & Camera Mapping, Multi-Pass Compositing, Color Grading & Look Development, Integration of CG with Live-Action, Shadow Matching & Reflections, Depth & Parallax Effects, Advanced Matte Painting.	**30**

IV	**Final Project & Industry Preparation** Project Planning & Asset Management, Shot Breakdown for Production, Using Shot Grid or Production Tracking Tools, Rendering & Optimization, Render Layering Techniques, GPU vs CPU Rendering, Render Passes & Post-Processing, Portfolio Development & Showreel Creation, Choosing & Editing Best Shots, Creating a Demo Reel for VFX Studios	**30**
	TOTAL	**120**

Textbooks:

- Producing VFX: A Guide to Managing the Process *by Charles Finance & Susan Zwerman*
- Filming the Fantastic: A Guide to Visual Effects Cinematography *by Mark Sawicki*

Reference Books:

- The Visual Effects Producer: Understanding the Art and Business of VFX *by Charles Finance & Susan Zwerman*
- Digital Compositing for Film and Video (4th Edition) *by Steve Wright*

Semester VII	
Major Course 4	UI/UX for Interactive Media
Subject Code	AVE092M714
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: This course provides students with essential skills in interface design and user experience research. It emphasizes user-centered design, enabling learners to create intuitive UI/UX solutions. Through wireframing, prototyping, and usability testing, students develop and refine functional designs, leading to a high-quality final prototype.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Recall** the principles of interface design and the different methods undertaken to study user experiences.	**BT 1**
CO 2	**Demonstrate** user-centered design principles to the creation of effective UI/UX solutions.	**BT 2**
CO 3	**Apply** wireframing, prototyping, and usability testing techniques to develop functional design outputs.	**BT 3**
CO 4	**Analyze** the designs to plan and execute a final quality output of the prototype.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction to User Interface**: UI history, User Centered Thinking, Context and Empathy, UI Design Process, User experience, Action History, UX processes and development	**30**
II	**Wire framing:** Design wireframes for a website or app based	**30**
III	**Prototyping:** Turn a wireframe into an interactive prototype, Usability Testing	**30**
IV	**Mobile Design:** Redesign an existing website with a mobile-first approach	**30**
	TOTAL	**120**

Textbooks:

- Do not Make Me Think: A Common-Sense Approach to Web Usability *by Steve Krug*
- *The Design of Everyday Things by Don Norman*

Reference Books:

- Hooked: How to Build Habit-Forming Products by Nir Eyal
- *Lean UX: Designing Great Products with Agile Teams by* Jeff Gothelf & Josh Seiden

Semester VII	
Minor Course 1	Introduction to Architectural Modelling
Subject Code	AVE092N711
Credit	4
L-T-P-C	0-0-8-4
Scheme of Evaluation	Practical

Course Objective: The objective of this course is to equip students with the fundamental skills required for architectural modeling, texturing, and rendering using **3ds Max**. Students will learn how to create detailed and realistic architectural models from various architectural styles, apply different textures and materials, and produce high-quality renders using industry-standard render engines.

Course Outcomes:

On successful completion of the course the students will be able to:		
CO Level	**Course Outcomes**	**Blooms Taxonomy Level**
CO 1	**Find** different forms of architecture to visualize the art form for better composition.	**BT 1**
CO 2	**Compare** the different era architecture into digital form.	**BT 2**
CO 3	**Apply** the knowledge of texture to beautify an architecture of different era.	**BT 3**
CO 4	**Examine** the appealing renders using models, texture, and lights.	**BT 4**

Course Outline:

Modules	Course Contents	Periods
I	**Introduction to Modelling**: Create different types of Prehistoric, Modern, Classical, Vernacular, Contemporary Architectures, Importance of architecture in different historical periods	**15**
II	**Types of Modifiers:** Extrude, Modifiers, Object Modifiers, Edit Modifiers, and World-Space Modifiers	**15**
III	**Materials & Texturing:** Baking of different Maps such as Diffuse Map, Metallic Map, Roughness Map, Transmission Map, Height Map, Displacement Map, and Sub Surface Scattering Map, Texturing software to finalize the look	**15**
IV	**Physically Based Rendering** Using Render Engine like Arnold, V-ray, Lumion to export the final Render	**15**
	TOTAL	**60**

Textbooks:

- Architectural Model Building: Tools, Techniques & Materials 1st Edition *by Roark T. Congdon*
- A Practical Study in the Discipline of Architectural Model making *by James Taylor-Foster*

Reference Books:

- The Theory of Architecture: Concepts, Themes, and Practices *by Paul-Alan Johnson*
- The Impact of Building Information Modelling *by Ray Crotty*